ENGLISH

Chapter 5

Light

Of all the science I've studied over the years— as a research neuroscientist, in my role hosting many of the world's most qualified doctors and scientists on my podcast, and in my own real-world experience—light, and the incredible influence it has on every aspect of our health, is the subject that has thrilled and surprised me the most. As a master's student I worked on the visual pathways that control circadian rhythms, and in graduate school and during my postdoc fellowship, and in my own lab, I researched visual circuits that convert light energy—photons, that is—into conscious visual perception, or what is commonly referred to as eyesight. My lab also spent a few years exploring the connections between our eyes and the cerebellum. Turns out, our eyes and our cerebellum—or "mini brain"— are in an ongoing dialogue that helps us orient in space. We Will get back to the cerebellum later when we discuss learning in chapter 6.

But as my lab explored the visual system, papers kept coming out showing how light either supports or damages the health of our cells. In some cases these findings were about how light affects the skin. In other cases, it was about how light impacts our health via our eyes in ways that are separate from eyesight. Given that so much of my training centered around our visual system, that research intrigued me. Not just because it was in my wheelhouse but also because of how actionable it was.

It could be applied easily to daily life, and impact health in potent ways. So I voraciously read every one of those studies. I also knew one of the pioneers in this field. Glen Jeffery, PhD, at the University College London, a former colleague of mine from the field of visual system development, began to study how particular types and sources of light either help or hurt human physiology. His lab also studied how light affected longevityu I connected with him periodically to get a sense of what was on the horizon.

While his is not the only lab working on the relationship between light and health, Glen is a leader in the formal study of this topic. We have him to thank for taking what were largely anecdotal observations put forth by niche online communities and testing them in controlled studies including clinical trials. Over those years, I was also spending more and more time indoors, under LED lights—both during the day and at night. Such is life in a lab. My colleagues all did the same.

So did my friends in medicine, tech, design, and, once smartphones hit the world full force in 2010, pretty much everyone else. I, and some of my more healthconscious colleagues, started to wonder if all that time indoors under the LED lights was part of the reason a lot of us were feeling so lousy. Now we know that it almost certainly was. The lack of full-spectrum

sunlight, and the use of LED light was no doubt directly damaging our bodies, inside and out, by impairing cellular health. Now of course, I can't be sure it was all due to indoor lighting, but at one point my hair started falling out, I was always tired in the afternoon, and my joints ached.

And I was exercising and eating well that light can be used to treat various diseases including tuberculosis of the skin, or lupus. Nowadays, photobiomodulation has been shown to be effective for treating acne, accelerating wound healing,5 and promoting hair growth,6 as well as reducing pain and even improving blood sugar regulation. Studies have also demonstrated the emcacy of photobiomodulation for offsetting age-related vision los* and for improving hormone health.JO That's not all; light has also been shown to improve heartu and GI tract healthJ2 and to help combat various autoimmune conditions.J3 And of course, there are countless positive effects of viewing sunlight on your circadian rhythm. As you recall from chapter 1, viewing low solar angle sunlight in the morning and again in the afternoon sets your sleepwake cycle correctly, increases your daytime mood, alertness and focus, and immune system function, and promotes better sleep at night. It is not exaggerating to say that light is medicine, especially sunlight.

And like any medicine, the effects of light depend on the type, intensity, duration, and timing of its delivery. For example, whereas sunlight in the morning and daytime is beneficial, viewing bright lights (especially from LED sources) too close to your bedtime spikes your cortisol levels, quashes your melatonin, and disrupts your blood sugar regulation (as discussed in Sleep Protocol 3). That affects your sleep that night and your energy the next morning. And certain wavelengths of light—in particular those emitted by most LED sources— may also accelerate cellular aging and promote disease progression by damaging cells in the retina, cornea, and brain, according to studies in animal models.L4 While all these effects of light might sound tricky to navigate, it's actually quite simple. You just need to know which types of light are beneficial to view, which are detrimental, and how and when to maximize one and minimize the other.

In this chapter, 1'11 teach you. What Is Light? First we need to discuss what light is, exactly. Obviously, light makes things visible. Any light source, be it the sun, a lamp, a candle flame, or the glow from a Phone screen makes things visible because it emits energy.

Light sources vary by type and arnount of energy; that's what makes them different from one another. Recall from chapter 1 that light is comprised of a broad range of wavelengths, from very short wavelengths to very long

ones. Some of these wavelengths are visible to humans—they make up what we call the visible spectrum of red, green, blue, and so on. Other wavelengths of light, such as ultraviolet and infrared, are invisible to us. Short wavelengths of light are, as their name suggests, waves of energy where the peaks and the troughs of the wave are very close together.

Long wavelengths of light are the opposite; the peaks and troughs of long wavelength light are relatively far apart. The sun emits an extremely broad range of short, medium, and long wavelengths. The entire visible spectrum is held inside sunlight. That's Why sunlight through a prism comes through as a rainbow on the other side. The prism slows down and bends the light, and each wavelength exits at a different angle, spreading the light into the full visible spectrum of red, orange, yellow, green, blue, indigo, and violet.

Green, blue, and purple light are shorter in wavelength compared to yellow, orange, and red light, which are longer. Sunlight (in the absence of a prism) appears white or pale yellow. The sky turns colorful at dawn and dusk because, as you may recall from chapter 1, when the sun is low in the sky, components of the atmosphere reduce the amount of shorter wavelengths reaching our eyes—a process called Rayleigh scattering —leaving us primarily seeing the longer wavelengths of red, orange, and yellow. The sun of course always emits the same wavelengths of energy regardless ofwhere it happens to be in the sky. Which visible wavelengths arrive at our eyes is what determines how the sun appears to us, and the effect it Will have on us.

Then there are the invisible wavelengths. Sunlight includes wavelengths shorter than blue or violet that are (unimaginatively) called ultraviolet (UV) light. We can't perceive UV light but it's there, radiating down on us all day. (Though interesting to note that bees can see UV, and squirrels can see it too, possibly to help them detect predators at night. Some species of flying squirrels actually have UV fluorescence that causes their bellies to glow a bubblegum pink.)L5 The sun also emits wavelengths longer than red light, appropriately called infrared (IR) light. We humans can't see IR, but we can feel it as warmth.

Though we have engineered technologies to see IR emissions—that's what night vision goggles do.

mutations in our cells by causing the formation of abnormal bonds in our DNA.J6 Wavelength and intensity of light are inversely related to one another. Put in practical terms, you can be exposed to a lot of long wavelength light, like red or IR light, without damaging your skin. It takes comparatively less exposure to short wavelength UV light to cause a

sunburn. But that doesn't mean short wavelength, high frequency light is inherently bad for us. Again, we come back to this core idea: Dosage matters.

We all need some UV light from the sun to reach our skin, where it sets off a cascade of reactions that results in the production of vitamin D, which is critical for calcium absorption, bone health, immune health, and more.UZ But too much UV exposure can lead to a sunburn, as I mentioned above, and cause premature aging of your skin, as well as DNA mutations that lead to certain skin cancersu-g And because long wavelength IR is so low energy (remember, wavelength and intensity are inversely related), we need relatively long sun exposure times to enjoy its benefits. The challenge is getting enough IR and NIR (infrared and near-infrared) light from the sun without also getting excessive UV exposure, but it can be done. There's a protocol for it. I'm aware that none of what I've shared with you thus far tells you exactly how light can change your cells. As wavelengths travel through space, in that up-and-down cycle, they hit various surfaces, big and small: a mountain range, a tree, this page, your hand.

These surfaces absorb or reflect wavelengths. But sometimes they pass right through. Light can pass right through your body, for example, and as it does it can change the molecular properties of the cells it encounters and the components within them, such as in the example of excessive UV-altering DNA structure of, say, skin cells. Here are two more examples, this time with light playing positive roles in cellular function: Studies suggest that red, NIR, and IR light can enhance the functioning of your mitochondria, the energy producing organelles in your cells.J-2 Data suggest that NIR and IR may also trigger the release of a gas called nitric oxide that acts on your vascular and lymphatic cells and causes them to dilate, allowing for more zo lymph fluid and blood to move through them. There are more examples in the protocols that follow.

You'll learn how to use light—in some cases sunlight, and in other cases artificial light of specific wavelengths—to improve your physical and mental health. Also here are recommendations for in-home lighting and light devices that emit or block specific wavelengths of light, which you can use to achieve specific benefits, such as increased morning and daytime energy levels, better glucose regulation, and even better eyesight. In chapter 1, we explored how light helps with sleep; now it's time to see its immediate effects on other important aspects of our health. As you read these protocols, remember this: While it might seem fantastical that light can do so many positive things for our brain and body, it actually makes perfect sense. Light is energy that the cells in our body can absorb and use.

The very first law of thermodynamics is that energy can't be destroyed or created de novo; it can only be converted from one form to another. So let's discuss the ways you can convert light energy into cellular energy, which leads to improvements in physical energy, vision, skin, mood, and heart health... and more. I hope this powerful lever Will no longer be overlooked. Note: If you are reading this bookfrom beginning to end, you'll realize as you encounter Light Protocols 1, 2, and 3 that these are essentially the same as protocols covered in chapter 1 and chapter 3. You may be thinking, "Does he have a memory issue, or think I have one?" The answer is: no and no.

The practices I suggest in this chapter are indeed found in earlier ones. But, because many people may read this book out of order, I wanted to be sure that these crucial protocols would not be missed. Moreover, the information that follows is not a carbon copy of the protocols in chapter 1 or chapter 3. I include new information about the mechanisms behind the "critical window" in the first hour after you wake up, whereby your SCN can amplify your morning cortisol levels... if and only ifyougive it the right stimulus. I also cover new information about viewing light at specific times to adjust your cortisol rhythm, so you can avoid what is commonly referred to as "burnout," but is often a mistiming in your daily cortisol peak.

I also include more options for indoor lighting—some were mentioned only briefiy in previous chapters; others are new. All that said, if you feel you are already versed enough in these first three protocols and most importantly, you are already doing them on a daily basis, _feelfree to skip to Light Protocol 4.

▸ View Bright Light in the First Hour After Waking to (Further) Increase Your Morning Cortisol Levels

WHAT TO DO

• Get morning sunlight in your eyes as soon as possible after waking. Ifthe sun hasn't risen yet, or is otherwise inaccessible, View a 10,000 lux artificial light. Either one Will significantly elevate your post-waking morning cortisol levels, which is highly beneficial if it occurs in thefirst hour ofyour day. • The timing Will vary based on when you wake up, and where you live. Refer to the information below to figure out what's best for you. Ten to thirty minutes is a good general rule, but even if you can only grab five, do it.

Some is better than none.

HOW IT WORKS

It is hard to overstate the positive impact of this protocol on your energy, mood, focus, stress poles in winter—which I'm guessing you don't— there is alvvays some sunlight during the daytime. So View it within an hour of waking. Let's keep going with those common questions. "Can I wear sunglasses?" It's best to View morning sunlight without them. Wearing corrective lenses such as eyeglasses or contact lenses is fine, even if they include UV protection.

Don't wear a brimmed hat, however. The neurons in your eyes that you need to stimulate here respond to light coming from above and in front of your eyes, and a brimmed hat Will block it. "Can I blink?" Yes! Never force yourselfto stare at sunlight, or any bright light for that matter. It can damage your eyes to do so.

People vary in their sensitivity to light, so if this is hard for you, squint and look to the area adjacent to the sun as needed. This shouldn't hurt. "Can I get my morning sunlight through a window?" While better than nothing, the glass used for most windows filters out too much of the wavelengths of light you need. Thus, go outside. Bring your morning coffee, tea, or a glass of water.

Bring your dog, kids, partner, or a friend along with you if you like. Yes, your dog Will benefit too. After all, the same circuits and mechanisms that underlie this protocol are present in humans and dogs (and horses, goats, and squirrels, as I've mentioned, and all diurnal mammals). And of course you can walk, jog, skip rope, or do calisthenics as you View bright light too. If you take an hour walk, or schedule a run or a yoga session outside, even better, as sunlight and exercise can synergize to increase your early morning cortisol levels.

Whatever you do, just don't spend the whole time looking down at your phone. That Will negate the effect you're seeking. "How long should I View morning sunlight to get the maximum effect?" As I mentioned, the answer depends on sunlight availability but a good general rule of thumb is this: On Clear mornings shoot for a minimum of ten minutes, on partially cloudy days, get a minimum of twenty minutes, and on very overcast days, thirty minutes. Ifyou View a 10,000 lux artificial light beforehand, for any of the reasons discussed already, you'll need less time viewing sunlight— about half as much—to get the same effect. It's hard to be more specific than this because sunlight availability is shaped by the time of day, time of year, weather, and location.

And, once more: "What if I wake up before sunrise, then what?" This is by far the most common question I hear. Unless you have powers that I am not aware of, you Will have to wait for the sun to come out to see sunlight. Get your inthe-first-hour-of-waking bright light by viewing a 10,000 lux artificial light. Ten thousand lux artificial light devices are not as good as sunlight, because they don't contain the same array and balance of wavelengths present in sunlight as discussed in the introduction to this chapter. But they Will elevate your morning cortisol levels, which is your goal in this protocol.21 They cost about $ 100 to $150.

Some are portable and suitable for travel. And, again, if you rely on a 10,000 lux light first thing in the morning, you'll Still want to get five to twenty minutes or so of natural sunlight exposure to your eyes once the sun comes out. Most people find that having a 10,000 lux light in their bathroom or kitchen works well for their morning routines. If you do use a 10,000 lux light, you needn't be obsessed with the viewing distance, nor do you have to worry about intermittently stepping or pivoting away from the light while preparing for your day. Just aim to be in front of it at a distance of about one to two feet for at least ten minutes total in the first hour after waking.

More is fine, but again people vary in their sensitivity to light and some find that after ten minutes of 10,000 lux light they feel too awake! Other people can handle more exposure. And if you have a question about my schedule: Because I typically wake up before the sun is out, I have a 10,000 lux light in my bathroom and my kitchen. I turn them on within a few minutes of waking and leave them on as I brush my teeth and prepare for my day. Once the sun is up, I go outside and View sunlight.

That way, on days when I need to commute to work early or exercise before the sun is up, I know I have already spiked my morning cortisol. And when I travel, I bring a small portable 10,000 lux unit and use it in the same way. It's a fantastic tool but it's not a replacement for sunlight. "Does viewing my computer screen or Phone work to replace morning sunlight or 10,000 lux light?" Alas, no. Your Phone screen is far too dim and far too narrow a spectrum to accomplish the effect here.

Diabolically, however, your Phone or computer screen is plenty bright enough to stimulate cortisol release if it's viewed at night. And that is bad. And your Phone or computer is plenty bright enough to suppress your melatonin levels. Also bad!

not advantageous if they occur in the afternoon, the evening, or at night, as I've mentioned, you actually want your cortisol levels to be very high in the

first hours of your day. Cortisol has gotten a bad rap. It's been labeled the stress hormone and indeed, this hormone made in your adrenal glands is released during times of stress. But cortisol is also released when you wake up each morning. More accurately, you wake up each morning because cortisol is released.

You read that right. Cortisol is the hormone that ensures you have the energy to get out of bed and lean into your daily activities. That is because cortisol's fundamental role is to cause the release of stored fuels in your liver and muscles, which shifts the state of your brain and body from sleep to one of physical and mental readiness. The technical term for this process, which you might remember from chapter 3, is called the cortisol awakening response (CAR). Cortisol is not just a stress hormone.

It's an energy-releasing hormone. (See Stress Control Protocol 5 if you'd like a reminder.) When we View cortisol in this way, we start to understand that it's not something to be avoided at all. Rather, it is something to harness and use to our advantage. Our body has designated pathways and mechanisms to do just that. It allows for what's called the healthy twenty-fourhour cortisol curve, which follows the pattern below: I Very Iow in the I to 2 hours prior to bedtime 2 Atits absolute Iowest in the first half of your night's sleep 3 Rising slowly in the second half of your night's sleep 4 Rising sharply in the 30 minutes prior to your desired wake-up time 5 Rising sharply even more in the first 1 to 2 hours after you wake up 6 Peaking about 30 minutes to 3 hours after waking 7 Gradually decreasing across the early and late afternoon 8 Repeat.. Countless studies now show that allowing this natural cycle to operate without disruption helps optimize our sleep, cognition, and energy levels as well as mood and immune system function.Z2 By contrast, altering this cycle such that evening or nighttime cortisol levels are too high (or morning cortisol levels are too low) can lead to problems in all these areas.

Studies of night shift workers have been crucial in uncovering these connections. A chronic disruption to the ideal cycle is referred to as a flattening of the cortisol curve. This flattening causes anxiety, sleep disturbances, low daytime energy, and mood disruptions as well as impaired immune resistance to viral, bacterial, 2.4 and other infections. Cortisol and Stress Keep in mind that just because we are framing cortisol as not purely a stress hormone does not mean it is not related to stress. It is.

Stress triggers the release of cortisol.25 If you experience stress at any point during the day or night, you can and should expect a spike in your

cortisol to occur. That's perfectly normal, healthy, and adaptive— because that spike Will provide you with the immediate physical and mental energy you need to deal with whatever stressor you might face. Stress-triggered cortisol levels typically return to baseline within about one hour or so. In fact, you can experience a lot of cortisol spikes above baseline per day (or night) before it ever becomes problematic for your health. I'm not suggesting you seek out stress, but as we discussed in chapter 3, humans evolved to deal with it in a variety of forms.

Raising kids, tramc, dificult people, the pressures of work, school, or athletics, the state of the world... many of these are indeed stressful at times, and cortisol makes sure you have the fuel to deal with each one. Cortisol spikes that occur shortly before sleep are where problems begin. That increase is often what prevents people from falling asleep, and what causes the classic stress-induced sleep disturbance of waking up at 3:00 am. and not falling back asleep. One night of that happening, no biggie. Two nights in a row?

Hard, but manageable. However, three or more nights of stress-induced sleep disturbances really start to take a toll on your mental and physical health. Cortisol and Light And now we get back to the protocol suggested here. I discussed how to deal with both acute and chronic stress in chapter 3. What we didn't discuss is that when you get your twenty-fourhour cortisol curve correct, it makes both kinds of stress much easier to deal with.

And to get your twenty-four-hour cortisol curve correct we need to... View morning sunlight (or 10,000 lux light) in the first hour after waking. It not only boosts your morning cortisol levels, but also helps ensure your nighttime cortisol is kept low. This has immediate benefits you can feel within minutes and that last all day, among them improved mood, focus, and energy. Think of it as the optimal start signal for your day.

Because from the perspective of your brain and body, that's exactly what it is. Here are the basics of how it works, plus some additional information pertinent to the twenty-four-hour cortisol cycle. Your cortisol level at any one moment, and your overall twenty-four-hour cortisol cycle, are controlled via the hypothalamic-pituitary-adrenal (HPA) axis, which we discussed in chapter 3. Neurons in an area of your hypothalamus called the P VN (paraventricular nucleus) extend wirelike projections into your pituitary, a gland that resembles two small marbles in a sock hanging out the base of your brain.2 One of those marbles sits in front of the Other, and this is called the anterior pituitary. Your PVN neurons release

corticotropin-releasing hormone (CRH) to cause your anterior pituitary cells to release adrenocorticotropic hormone (ACTH) into your bloodstream.

Next, the ACTH travels to your adrenal glands and stimulates the release of cortisol into your bloodstream. Voila! Cortisol then causes energy to be released from various organs and cells in your body. Cortisol does many other things, but one more that's relevant here: Unlike adrenaline, which is also released from your adrenal glands and increases your energy level, cortisol can cross the blood-brain barrier. That means it has action on the body, such as mobilizing glucose for fuel, but also can increase some of the immediate benefits of stimulating cortisol in the hour after waking, but perhaps by now you can also identify another: If you do, you not only get enormous benefits in energy, focus, and mood right away, you also kickstart that crucial negative feedback loop that ensures that your levels of cortisol Will be dropping as you head into the afternoon and evening.

Your cortisol levels Will be very low as you progress into sleep. That's how all the pieces come together: In that special window in the first hour after you wake up, when your SCN can also stimulate a significant increase in your cortisol levels, you can bypass the negative feedback loop and stimulate the release of more cortisol even though its levels are already high. In fact, your SCN does so both through the HPA pathway and via a separate, parallel pathway involving direct activation of your sympathetic nervous system.28 We can refer to that second pathway as the MLIC pathway (for morning-light-inducedcortisol pathway). And, as the name indicates, the way to activate the MLIC pathway is by viewing bright light. As a final note, researchers have studied the use of bright artificial light in the first hour of the day not just for this, but as a therapeutic intervention for seasonal affective disorder (SAD), a diagnosable mood disorder characterized by depression that fluctuates with the seasons, worsening in the colder, darker months.22 Rates of this condition are particularly high among people who have major depression or bipolar disorder.

In one study, people diagnosed with seasonal affective disorder were treated with twenty, forty, or Sixty minutes of 10,000 lux light in the morning. Even the shortest exposure was suficient to relieve depression among the participants, according to a self-reported questionnaire. Other studies have shown similar resultsn Newer studies have shown that reducing the amount of bright light reaching your eyes in the final one to two hours of your day leads to improvements in mood and sleep, and to various mental and physical performance benefits as well.3-Z In the last ten years, it's become Clear that far more people are suffering from SAD or

SADlike symptoms due to device usage in the early and late hours of the day. In chapter 1, we discussed the severe negative effects of shift work on long-term health, and the fact that nowadays most adults and kids—most (!)—qualify as shift workers due to this kind of device usage. How can that be?

Well, remember—even the brightest device screens are too dim to trigger meaningful cortisol increases in the first hour of your day, and even the dimmest screens are more than bright enough to trigger cortisol increases at night. That is, unless you filter out the short wavelengths reaching your eyes—which we talk about in the next protocol.

COMMON QUESTIONS

Does it matter if this light hits my skin or are the eyes all that count here? This protocol is concerned with viewing light only. Later protocols in this chapter Will address the need for sunlight on skin and how to balance that exposure with cautions about sun damage.

LIGHT PROTOCOL View Sunlight in the Late Afternoon and/or Early Evening to Help Buffer Against Negative Effects of Artificial Light at Night

WHAT TO DO

• View sunlight in the afternoon and early evening, for five to thirty minutes. Follow the same general guidelines as for Light Protocol 1. Simply go outside, take off your sunglasses (eyeglasses and contacts are fine), and face in the direction of the setting sun. Don't wear a brimmed hat. And again, don't stare at the sun; blink as needed. • Do not replace late afternoon/evening sunlight viewing with a 10,000 lux artificial light.

It's too bright, and the wavelengths these devices emit can mess up your sleep-wake cycle.

HOW IT WORKS

As discussed in Sleep Protocol 2, melatonin, which facilitates falling and staying asleep, begins to increase in the late afternoon and early it's bright, and 10,000 lux is bright—will alter your sleep-wake cycle. You'll notice, just as in the previous protocol, that I'm not suggesting specific times to View late afternoon or early evening sunlight. Again, that is because schedules vary from person to person, and the time of day when the sun sets varies according to your location and the time of year. Luckily, you can get the desired effect without being overly precise about the timing. Just aim to

View sunlight at some point after the sun is directly overhead and before it sets.

This doesn't mean that you should avoid sunlight or artificial light during other parts of the day. Researchers from Rush University ran a small study where, for one week, people experienced bright light exposure for up to five hours by combining outdoor exposure to sunlight and indoor exposure to 10,000 lux light during the day.35 They called it "the bright week." The following week, the light exposure was dimmer; people spent two and a half hours outdoors at most, and wore light-obscuring goggles when they did. That was "the dim week." The researchers found that light-induced melatonin suppression at night was greater during the dim week than during the bright week (though some participants showed greater differences than others). Bright light exposure during the day partially prevented light-induced melatonin suppression at night. Seems paradoxical, given that light suppresses melatonin, right?

Yes, until we look at another mechanism at work here. Another small study from the Netherlands measured melatonin levels, body temperature, and skin blood flow (all markers of circadian rhythmicity) of twelve women and noted how sleepy they reported feeling, according to when they viewed light in the evening and at night.36 They divided subjects into three groups. One group viewed dim light at night. Another group also viewed dim light at night but also viewed bright lights beforehand, late in the evening. The third group viewed bright light in the early evening, and then dim lights at night.

All three groups saw some reduction in melatonin, which is not surprising given that light viewing suppresses melatonin production any time of day or night. But the third group —the group that saw early evening bright light plus late-night dim light—had a less dramatic melatonin drop. This again supports the idea that early evening light exposure can partially inoculate us against the negative effects of bright light at night. These studies may be small, but they point to the importance of viewing sunlight in the late afternoon and early evening, and avoiding bright light exposure at night. It's worth noting that it may be especially important for adolescents to avoid bright light at night, as one study showed that early evening sunlight exposure did not protect them against the melatonin-reducing effects of bright light as it did for adults.3-Z This may also mean that adolescents are particularly sensitive to the negative effects of bright artificial lights and screens at night.

The reasons to try to offset the negative effects of bright light at night are not just related to sleep. Animal studies have shown that bright light exposure to the eyes late at night activates pro-depression circuitries.3.3 And as we've already covered, human studies have shown strong links between nighttime bright light exposure, cortisol increases, melatonin suppression, and mood disturbances. If you or your Child happens to View bright light in the hours before sleep now and again, perhaps from a movie or a night out, that's no cause for concern. If you make a habit of keeping bright artificial lights on late at night, however, and you notice your moods are shifting for the worse, that could be why. As you're probably realizing, modern life—with all its lighting options and screens everywhere —is not oriented around daylight fluctuations in the way it once was.

Humans used to wake up and go to sleep with the sun. We had no choice but to be attuned to the solar cycle. The advent of artificial lighting changed everything, and the shift from incandescent to lower cost LED lighting further exacerbated the issues. In fact, in the last ten years, the arnount and type of artificial light exposure humans experience has changed so dramatically that we can honestly say that, as a species, our whole relationship to light has been altered. Yet our biology hasn't changed one bit in thousands of years.

Hence this chapter.

COMMON QUESTIONS

How do I make this work if l'm stuck in the office all day? There is no need to be obsessive about this— it doesn't have to interfere with your life. Still, by making some concerted effort toward viewing sunlight (or a 10,000 lux lamp) in the mornings, and sunlight in the late afternoons or early evenings, and avoiding bright light at night, you can dramatically improve your mood and energy levels, metabolism, and sleep—which of course bolsters all the rest. So try to get outdoors at these times if you can. Better yet, take a walk in the direction of the sun.

Or even just make time to turn toward the sun and pause (and not look at your phone) for a few minutes in the parking lot before leaving work.

▸ Avoid Viewing Bright Light One Hour Before Your Bedtime

WHAT TO DO

• In the one hour before your bedtime, use only as much artificial light as necessary to safely carry out your nighttime activities. That recommendation holds until you wake up the next morning to begin your day. • Be especially mindful to avoid viewing bright short wavelength light at this time. Avoid not just lights that appear blue, but also bright "white" LED lights, fluorescent lights, and screens, because they emit a lot of short wavelengths that stimulate cortisol release and inhibit melatonin. • If you can't avoid viewing bright short wavelength light, consider wearing short wavelength light blocking lenses. Not simply "blue blockers," but these specific lenses. For anyone seeking an easier transition into sleep and better overall sleep, these glasses can be helpful.

HOW IT WORKS

In the first hour of your day, it takes a lot ofbright light to stimulate cortisol release and activate before sleep' stuff? No one lives this way! Why can't you just give me a clock time?" I've said this before, and 1'11 say it again here: People follow different sleep-wake schedules, with some sleeping from 8:00 pm. to 4:00 am. and others sleeping from 11:00 pm. to 7:00 am. Still others work the night shift. Variation in people's schedules makes it impossible for me to say, "Dim the lights and avoid viewing LED screens between 10:00 pm. and 5:00 am." or even "View bright light within an hour of sunrise." Also, the length of days varies according to location and time ofyear, and things like daylight further complicate matters.

So I suggest you frame your sleep-wake schedule according to four phases. Once you try this approach to your days, and use light (or avoid it) accordingly, I'm certain it Will improve your health and well-being. It's worth noting again that the clock times corresponding to the four phases Will change if you were to, say, sleep in an extra hour or two, or wake up an hour earlier than is typical, or go to sleep an hour earlier or later. Not a problem! When it comes to the effects of light on cortisol and melatonin, we care about your biology, not an alarm.

Start phase 1, for example, when you get up to begin your day. The light prescription for each phase does not change. This ought to bring a sense of control over your time—after all, no one wants to have to take care of their health strictly by the clock. Work, school, and other commitments have us doing that already. In case any of this is confusing, here is how I use this system to structure my days (and nights).

My Phase 1 (as of writing): I woke up at 7:00 am. I usually wake up at 5:45 or 6:00, but I was a bit more tired than usual from a long week of work and

travel and I forgot to set my alarm. My sleep tracker told me that I fell asleep last night at about 10:15 pm. At 7:00 am., I went for a walk outside for about twenty minutes, without a hat or sunglasses, and walked east in the direction of the sun. It was light out but overcast, with no hint even of the sun's outline in the sky.

But because it was fairly bright, I knew I was getting a decent amount of sunlight in my eyes, and therefore stimulating cortisol release. I was not at all concerned that I had somehow missed the window to use sunlight to boost my cortisol because my usual wake-up time is about 5:45. I know it's about getting light in my eyes in the first hour after waking, not at a specific clock time, that matters. I returned from my walk and, just to be sure that I spiked my morning cortisol suficiently, I flipped on a 10,000 lux artificial light that I keep in my kitchen and kept it on while I went about my morning activities. Today was my day off from exercise (yesterday was leg day and generally the day after leg day is a day off from exercise for me; see Exercise Protocol 2).

From 8:00 am. until 11:00 am., I worked (podcast preparation, emails, and calls) at my desk with the overhead indoor lights on their brightest setting, and I sat near a window. I managed to get a few minutes outside during one of my work calls, and I paced while on the call, which allowed me to get a few extra steps in and a bit more sunlight. Around 11 am., I paused work to eat lunch, and then went back to work at my desk until about 2:00 pm. On a less busy day, I might have taken my lunch outside. My Phase 2: Given that I woke up at 7:00 am., this phase began Six hours later at 1:00 pm., and sure enough I started to feel a little sleepy around that time.

I almost always experience a dip in energy about Six hours after I wake up, even after a great night's sleep. Some people don't, but I do. I know it correlates, among other things, to the decline in cortisol that is normal, healthy, and expected at this time. I know better than to push through this dip. I could take a prescription stimulant or slam an energy drink or two at that time, but why?

I know there are other tools that won't mess with my sleep. So from 2:00 to 2:30 1 did a non-sleep deep rest (NSDR) session, which I described in chapters 1 and 3, and I emerged from it feeling refreshed. I drank some water and had a half cup of strong coffee and then dove back into work until about 5:00 pm. (I can drink caffeine up until about 2:00 to 2:30 pm. with no negative effect on my sleep, but for those more caffeine sensitive, take care.) My Phase 3: Phase 3 is from ten hours until about fourteen hours after waking, so for today, that meant from 5:00 to 9:00 pm. I used

this time for more work. I also took a brief twenty-minute walk outside, then prepared and ate dinner.

Then I started to wind down for the night. I dim the lights in my home to less than full blast starting in phase 3, and I recently started using the medium and long wavelength only setting in the evenings on lights that switch from full spectrum to medium and long wavelength emitting only. (See Sleep Protocol 3 for more on these bulbs.) They appear yellowish-orange rather than white. Several companies such as Bon Charge and Ra Optics now make LED bulbs that switch from full spectrum to medium and long wavelength emitting only (without having to program them for a specific time). You just flick the standard wall-mounted light switch. I love this simple technology because it's so aligned with what our system needs, and it's easy to use with no Wi-Fi or an app.

And because they are LEDS, they last a very long time and are cost-emcient. My Phase 4: This phase spans from the period starting fourteen hours after waking until the next morning when the new day starts. Today for me that meant from 9:00 pm. on. For some time now, I have made it a practice to dim the lights in the final hours before I go to sleep. As I mentioned earlier in this chapter, people vary in their visual sensitivity to light.

I am one of those who can be in front of a screen right up until I fall asleep. I have no trouble falling asleep, but it tends to make me wake up three to four hours later. When I started dimming the lights before bedtime, those mid-sleep awakenings all but stopped. I also switched to using long-wavelength-only light bulbs (red light) in the one to two hours before my bedtime. I Still might wake up once in the middle of the night to use the bathroom but I have no trouble falling back asleep, and both my device-measured sleep scores and my subjective You can go further Still and use bulbs without short wavelengths, which thus appear yelloworange, or even go with bulbs that emit purely long wavelengths, and thus appear red.

Keep in mind this is not red light therapy. Red light therapy (explained later in this chapter) can be very beneficial for certain aspects of health but is altogether different from what we are discussing here. Medium-wavelength-only bulbs are relatively inexpensive and provide a nice yellow-amber glow. Admittedly, using long-wavelength-only bulbs Will make your home look a little space age, but they are a great tool to help you sleep better. Either way, using dim, non-overhead lights without short wavelengths in the last one to two hours before bedtime reduces the

activation of the retinal neurons that signal your brain to suppress melatonin and boost cortisol.4Z As mentioned above, several companies have conveniently engineered bulbs that are fullspectrum-emitting (white) for the daytime, and with a simple flick of the light switch, become medium and long wavelength only emitting (yellow-amber or red).

Incandescent versions ofthese exist too and are best, but they use more energy than LEDS, don't last nearly as long, and are more costly. They are also a bit hard to find. • Wear eyeglasses that filter out all short wavelength light. This approach is very convenient ifyou don't want to change the lights in your home, or if you don't have control over the lighting conditions in a certain environment, such as on planes, in hotels or airports, out with friends, et cetera. I designed the ROKA Wind Down glasses for use during the final hours before sleep, and I use them at home and when I travel. They differ from traditional blue blockers because they prevent all short wavelength light, not just blue light, from reaching your eyes. (Don't wear them while driving because they make anything that is blue or green much harder to see or even invisible, and that could cause an accident.) • Don't worry about moonlight.

You may be wondering if bright moonlight or even a roaring fire Will stimulate the wakefulness pathways in your brain. The answer is, perhaps surprisingly, no. Although these are concentrated light sources, a bright full moon is less than 1 lux and while candles, fireplaces, and bonfires mostly emit medium and short wavelength light, it doesn't radiate very far. Ifyou have ever been camping and turned around from a roaring campfire, you've noticed that as soon as you are just a few paces away from it, you need a flashlight in order to see. We discussed the physics ofthis in chapter 1.

So while fire and candles are great nighttime light sources from a cortisol-limiting and melatonin-preserving perspective, they are not very practical. Of course, ifyou'd like to read by candlelight or gather around a fire with friends, great. Just make sure the flames are extinguished when you're finished. Be careful not to start a fire. You may be wondering if all this attention to avoiding bright light in the hours before sleep is silly, neurotic biohacking.

After all, you can find studies showing that blocking blue light improves people's sleep to varying degrees, and you may be one of those people who can pass out with the TV or computer Still at full brightness right in front of you. Indeed, there is considerable variation in the degree to which short wavelength light exposure alters people's sleep.44 That said, having even

just a dim 100 lux overhead light on while you sleep can alter the next-morning glucose levels.45 And that's while sleeping, with eyes closed. The point is, altered sleep is not the only reason to avoid bright light in phase 4. Nighttime bright light exposure increases blood pressure46 and heart rate, and has been linked to weight gain and even a risk of obesity.4Z Studies have found nighttime bright light to negatively impact next-day cognitive functiono and mood42 independent of bright light's negative effects on sleep. None of this is surprising to biologists who have studied the so-called non-image-forming visual system or the many opsins other than melanopsin that are present in our skin and other tissues for the last three decades.50 Myself included.

We know, for instance, that the retinal neurons that respond to bright short wavelength light connect not only to your SCN but also to areas of your hypothalamus, such as the supraoptic nucleus, that are directly involved in sleep regulation. (I know, supraoptic sounds like suprachiasmatic but please don't blame me, I didn't name either one.) Your eyes also connect to your thalamus's perihabenular nucleus, which has been nicknamed the anti-reward or disappointment center, 1 givenit actively inhibits dopamine release and is thought to increase including age, sex, physical activity, smoking, drinking, and cardiovascular health. There's more. The analysis zoomed in on nighttime light exposure, grouping it into half hour increments and matching those increments with mortality data. The brightest lighting between 1:00 am. and 6:00 am. carried a risk of death by any cause that was 11 to 39 percent higher compared to the darkest lighting during this stretch of time. The risk of death due to cardiometabolic issues (heart attack, stroke, diabetes, nonalcoholic fatty liver disease) was 20 to 67 percent higher for the brightest lighting compared to the darkest lighting during this stretch of the night.

The time frame here differs slightly from the one described in this protocol, but the exact times aren't the point; the point is that viewing bright light at night is disruptive to your immediate and long-term health. It's true that this was a relatively brief study, using just a week's worth of light exposure data. However, about three thousand participants repeated the study up to four times and the light exposure patterns were the same. Several other studies have found similar links.5.3 And the sheer volume of light exposure data in this one makes a persuasive case. As the researchers wrote in the 2024 publication of their study in the Proceedings of the National Academy of Sciences, "Across the general population, avoiding night light and seeking daylight may lead to reduction in disease burden, especially cardiometabolic diseases, and may increase longevity." In another large

study, published in Nature Mental Health in 2023, researchers at the University of Manchester cross-referenced data on mental health with data on light, sleep, and physical activity from 86, 722 individuals.54 Their analysis revealed a link between greater nighttime light exposure and higher risk for several mental health issues, including major depressive disorder, anxiety, post-traumatic stress disorder, and self-harm behavior.

Greater daytime light exposure correlated with a reduced risk of these issues. They also found that people who experienced darker nights had fewer mental health symptoms, an effect that occurred independent of daytime light exposure. That is, bright-light-filled mornings and days and dark nights independently and additively promoted better mental health. These large datasets deliver a Clear and important message, one that typically goes unmentioned: Keeping the lights off during the late hours of your night is not just about getting great sleep; it is fundamentally important for your health and well-being.

▸ Expose Your Upper and Lower Body Skin to Sunlight During the Day

WHAT TO DO

• Get twenty to thirty minutes of outdoor sunlight exposure to your skin at least three times per week, and ideally every day. Wear the minimum amount of clothing that allows you to be comfortable and that is appropriate for your environment, without covering your entire body (for example, shirtless with shorts, or shorts and a tank top, or a swimsuit. In some cases, jeans and a short-sleeved T-shirt may be what's called for). It is best that you do not wear a hat during this time. I discuss cold weather options below.

HOW IT WORKS

There is no substitute for sunlight. You could turn on every light in your home in the morning and daytime, make sure they are full-spectrum incandescent bulbs or wavelength-balanced full this, especially given that we all have heard how damaging UV light is for our skin. Yet, we need to embrace what is special and beneficial about sunlight and incorporate it into our daily lives. And no, you do not have to live in California, or run around shirtless or in a bathing suit outside. You simply need to know how the energy present in sunlight impacts every cell in your body: your brain, hair, skin, liver, heart, your hormones, neurotransmitters, immune system, and on and on.

And so, on to the science. Sunlight is the absolute best source of long wavelength light, from the red end of the visible spectrum into non-visible near-infrared and IR wavelengths. It is dificult to overstate how important these longer wavelengths are for our health, in particular for our mitochondria, the energy producing organelles within our cells. (1'11 cover the uses of LWL devices that purely emit red, NIR, and IR light in the next protocol.) Sunlight contains short wavelength UV light, which as we discussed earlier in this chapter, triggers the production of vitamin D in the skin. Although this vitamin (which also acts as a hormone) is available in some foods, supplements can replace sunlight as a stimulus for vitamin D,55 yet our bodies evolved to use sun-induced vitamin D in multiple ways that supplements may not mimic. For example, vitamin D is important for hormone health and the reproductive system.

A multi-institution study on mice and humans looked at the effects specifically of UVB (the ultraviolet wavelength that penetrates the skin) on reproductive-related hormones and behavior.56 They found that mice exposed to UVB light for one hour per day showed enhanced mating behavior and physiology—greater attraction for one another, larger ovaries, and more testosterone—in both the male and female mice. Moreover, male mice showed a greater attraction to UVB-treated female mice, likely due to how elevated testosterone and estrogen levels in females impacted their appearance and scent, making them more attractive. More interestingly, the researchers saw similar patterns in people. According to their findings, published in Cell Reports in 2021, participants reported increased romantic interest in others following time in the sun. The data also showed heightened testosterone levels linked to this exposure in both men and women.

Testosterone and estrogen levels—both of which impact romantic desire and feelings of well-being in men and women—were highest in the longer days of spring and summer relative to shorter winter and fall days. This effect appears to relate to a molecule called p53 expressed in a type of skin cell called keratinocytes. Mice in which p53 had been removed did not exhibit the same changes following UVB exposure. This detail emphasizes the importance of having sunlight reach not only our eyes, but also our skin. Here's another: UV light from sunlight also reduces physical pain.

In 2017, a group of researchers reported a study in which they exposed mice to UVB (midrange ultraviolet light, abundant in sunlight) radiation and then tested various parameters relating to the HPA axis. This brain pathway is the core mechanism behind our stress response (as discussed in chapters 1 and 3).5-g They found a link between UVB radiation and the

stimulation of corticotropinreleasing hormone, which helps us cope with social stress,52 and endogenous beta endorphins, a type of opioid our body naturally produces in response to psychological distress.60 Although this work focused on mice, the results point toward a potentially crucial and unappreciated role for sunlight, in particular UV light—which most people generally think of as the aspect of sunlight to be avoided. Excessive UV light exposure Will cause sunburn and increase your risk for certain forms of skin cancer. So, while UV light from the sun has many benefits, it is Still important to avoid too much exposure and sunburn. A Word About Sunscreen Of course, no one wants a sunburn.

One's risk of sunburn and skin cancer varies depending on factors such as skin tone and inherited susceptibility to skin cancer. All dermatologists agree that any amount of sunburn is bad for us, so bear that in mind. And all agree that excessive exposure to sunlight leads to more rapid aging of the skin too. Excessive skin exposure to UV light, in particular, mutates our skin cell DNA and destroys keratinocytes, which give skin its suppleness. So the takeaway message is to avoid sunburn at all costs.

I don't burn easily but if I am going to be outside at a time of day when the UV index is high, I either rely on a physical barrier— hat and long sleeves—or I apply a mineral-based sunscreen that uses zinc oxide (not nano zinc) as the only active ingredient. People debate whether looking at mortality rates among 29,518 women found that greater sun exposure correlated with lower all-cause risk of death.6J In fact, people who spent the least amount of time in the sun had nearly double the mortality risk during the twenty years of the study compared to those who spent the most time in the sun. Obviously, there are likely to be many factors at play here: More time outside usually means more fresh air, movement, social interaction, et cetera, not just sunlight. Yet the data correlating different amounts of sunlight exposure to health and longevity all seem to point in the same direction: Getting sunlight on your skin is beneficial. Interestingly, that work followed an earlier study of nearly forty thousand women that found a reduction in mortality linked to exposure to ultraviolet light coming directly from the sun, and yet an increase in mortality linked to ultraviolet light from artificial sources.

Why would UV light from sunlight help and yet the (seemingly) same UV light from artificial sources harm us? One idea is that because sunlight has a balance of short, medium, and long wavelengths—each at particular energies and with their own unique health-promoting effects —they synergize. UV without adequate longer wavelengths—such as what comes from most LEDs—is thought to be detrimental to our cells.6.3 1'11 dive

more into the ways that different wavelengths of light impact our cells in the next protocol, but for now here are some crucial things to know: No single wavelength of light is "bad" or "good." Longer wavelengths of light are absorbed by the water in our mitochondria and help "charge" the mitochondria like batteries, making them more eficient at producing ATP and possibly even helping cells to make more mitochondria. Longer wavelength light also helps offset the negative effects of reactive oxygen species (ROS)—a normal by-product of cellular function that, if in excess, damages our cells.65 ROS are thought to be a major factor in the reduced energy levels and organ dysfunction we experience with aging.66 Long wavelength light can pass through our skin, eyelids, our ribs, and even our skull to reach our brain and other internal organs, and so helps to promote their health, largely by improving mitochondrial function as described above.6 How wild is that?! Sunlight also contains abundant amounts of medium wavelength light—yellows and oranges, or in technical terms about 500 to 620 nanometer light, which has been shown to reduce headache pain, stimulate blood flow, and reduce inflammation.

Artificial lights of course contain some ofthese wavelengths but not in the same balance as is present in sunlight. Incandescent bulbs are better than LEDs in this respect, but nowadays LEDs and fluorescent lights are almost exclusively used for indoor lighting because they cost less, use less energy, and last longer. Most LEDs are enriched with short wavelengths which, unless balanced out by long wavelength light, have been shown to increase ROS and damage mitochondria. Some LEDs and fluorescent lights even emit small amounts of UV.ZO This all becomes a problem for cellular health when we are under these lights all day and not getting the protective effects of long wavelength light or enough sunlight. It's rather incredible that while so many people are afraid of UV from sunlight, they may be unaware how the many hours of daily (and nightly) exposure to short wavelength light can damage our health.

The remedy: balanced wavelength exposure from sunlight, done safely of course. The protocols in this chapter represent the key times and ways to do just that. The other remedy: incandescent rather than LED indoor lighting, and limiting your short wavelength light exposure after nightfall. Still not convinced that light is medicine? I mentioned above how researchers found that UV light from sunlight reduces physical pain, and there are studies of humans experiencing similar effects.

In a study of eighty-nine patients who'd undergone spinal surgery, those who stayed on the bright side of the hospital, and were therefore exposed to 46 percent more intense sunlight than those staying on the dim side of

the hospital, reported less pain-related stress and a bit less pain, and required 22 percent less pain medicationn Similar results have been reported 7.2 in the relief of fibromyalgia among women and in a small study of US military veterans, chronic lower back pain. Preliminary research indicates that the underlying mechanism here brings us back to the intrinsically photosensitive retinal ganglion cells, which don't just connect to the SCN but also communicate light information (through a relay station) to the periaqueductal gray area, a region that is crucial to pain modulation. And, as discussed in other protocols in this chapter, regular, consistent light exposure also enhances mood.Z5 Data suggest that individuals exposed to moderate arnounts of UVB—which is if you can't get outside into direct sun, do what you can. Sometimes you can only get time out on a covered balcony. Every little bit counts.

It's like getting your daily steps in. You need your daily light exposure too. As a final note on this: If you live in a region where you just can't get much sunlight exposure to your skin for many days of the year, you would be very wise to get a 10,000 lux artificial light (any brand Will do), and a red and IR incandescent light. You might also use bulbs in your lighting fixtures that switch from bright, balanced full spectrum to medium and long wavelengths only simply by using the light switch.

▸ Expose Your Skin and Eyes to Long Wavelength Light (Red, Near-lR, and Infrared)

WHAT TO DO

• Use a long-wavelength-light-emitting (LWL) device for ten to thirty minutes, three to seven days per week. For general health effects (energy, skin appearance, exercise recovery, etc.), the goal is to illuminate the largest area of your skin, face, and head that you can. This Will obviously depend on what LWL device you have access to and the amount of time you can devote to this.

HOW IT WORKS

Red light therapy has become a popular health trend. Red light saunas, red light panels, red light beds, masks, even red-light helmets to promote hair growth. the list goes on. Despite their space age look, this is a trend with genuine scientific support behind it. Several leading dermatology journals including JAMA Dermatology have published in-depth reviews on the use of red light pass through thin clothing such as T-shirts and through your torso.$l That might sound scary but remember that long wavelength light

is low energy non-ionizing radiation, so it is safe, and all evidence points to long wavelength light having a protective and a health-promoting effect on your cells. Put simply, exposing your skin to long wavelength light in this way is a good thing.

The main reason? It promotes mitochondrial health. Mitochondrial health is a term used often these days, so before going any further, let's further define what this means. As discussed several times already in this book, cellular energy is generated in the form of a chemical called adenosine triphosphate, or ATP. Mitochondria are crucial for ATP production, as well as for making our hormones and neurotransmitters.

Mitochondria contain an enzyme called cytochrome c oxidase, and it can absorb lightg-Z In that way, cytochrome c oxidase is similar to the photopigments in your eye that absorb specific wavelengths of light, and that kick off the cascade of events in your eyes and brain that lead to sight. The difference is that cytochrome c oxidase is inside the cells of your body and brain, and its job is to support the formation of ATP for your cells to use. The way it does that is by moving electrons from one protein to the next inside the mitochondria. (Briefly: Like any compound, proteins are made of molecules; molecules are made of atoms bonded together; atoms are made of subatomic particles including protons, which are positively charged, electrons, which are negatively charged, and neutrons, which carry no charge.) That process releases energy that moves protons, also in the mitochondria, and because protons carry an electrical charge, it sort of works like a battery. The business of moving protons in mitochondria isn't just about making energy; it takes energy too. The first law of thermodynamics is that energy can't be created or destroyed, it can only be transformed.

Long wavelength light energy from the sun that is absorbed by cytochrome c oxidase is transformed by your mitochondria to help create ATP. So, in a loose sense, we are solar powered. The absorption of long wavelength light by cytochrome c oxidase also causes nitric oxide —a chemical that can impair mitochondrial function—to dislodge from the organelle, which then allows cytochrome c oxidase to use oxygen more eficiently. That enhances something called mitochondrial respiration, a process critical for converting nutrients in our blood into ATP Barn! Long wavelength light can enhance cellular ATP production through at least two proposed mechanisms.

Long wavelength light also may support mitochondrial energy production through a third mechanism involving the increased action of intracellular

melatonin, which can help prevent oxidative damage.8.4 Until now we have discussed melatonin only as a hormone produced by the pineal gland that promotes sleepiness. However, many cells throughout our body produce melatonin. Long wavelength light may allow intracellular melatonin to act more effectively as an antioxidant, helping to repair some of the oxidative damage that is a normal consequence of cellular metabolism. One potential benefit of this pathway is that it may reduce some of the harm caused by excessive UV light exposure. As we discuss next, the implications of that are, literally, visible.

How It Affects Your Skin Long wavelength light can promote the youthful look of skin, reduce acne, and accelerate wound healingg-6 Studies have also investigated long wavelength light as a tool to treat wrinkles, scars, burns, and other causes of discoloration of the skin. In one study of ninety-three patients, administration of rapid pulses of 590-nanometer light was linked to reduced signs of aging. In another study by the same research group, ninety patients were given pulses of 590-nanometer light in eight treatrnents over four weeksg-g The investigators took digital images of the patients at the start of the study and over the following twelve months and found reduced signs of aging in the vast majority of participants. They also evaluated tissue samples from ten of the patients and found increased collagen, a protein that provides structure and elasticity to skin, in all the subjects who received this therapy. The wavelength measuring 590 nanometers is in the orange-red range of the visible spectrum and, as noted above, longer wavelengths (600—850 nm) can pass even deeper into the skin layers.

Indeed, investigations found even more impressive reductions in wrinkles following treatment with 633-nanometer (red) light combined with 830nanometer light, which falls in the near-infrared range There also are benefits to combining long wavelength light with short wavelength light, only light sources like conventional LED lights lowers your metabolism, even going so far as to argue that obesity is on the rise because of short wavelength light overuse. While that might be the case, that is not what this study shows. It is reasonable, however, to conclude on the basis of this study that eating outside in the sunlight or even in the shade on a sunny day where long wavelength light is abundant, or eating near a LWL device, may be beneficial for those seeking to improve their blood sugar regulation. Getting exercise and proper nutrition per the recommendations in chapters 2 and 4 are the major factors for healthy blood sugar regulation. Nevertheless, long wavelength light can certainly be used to augment proper exercise and nutrition, and it may be a particularly useful

tool for regulating blood sugar and metabolism when exercise is restricted due to injury, weather, travel, and the like.

How It Affects Your Eyesight The effects go on. Long wavelength light has been helpful in preserving eye health too. First, a little bit about your eyes: The photoreceptor cells in your eyes consume more energy than any other cell in your brain or body.93 Think about that for a second: more than any other cell, meaning more than all the rest of your brain cells, your liver, even your muscle cells... and that is true even if you were to run a marathon and use those muscle cells, every single day! Your photoreceptors are almost always active, even in the dark. The point is your photoreceptors produce and use a ton of energy around the clock.

The energy they draw on comes in the form of ATP, and ATP production, as you now know, depends on mitochondria. Not surprisingly, your photoreceptors are packed with mitochondria94 and they produce a ton of ATP, but with all that ATP production comes a high amount of free radicals, and as I mentioned earlier in this protocol, those impose stress on the cells, which over time damages the mitochondria. Exposure of the eyes to long wavelength light —the red, near-infrared, and infrared end of the spectrum—both reduces the presence of free radicals95 and helps stimulate the production of ATP by the mechanisms discussed above. So it makes sense that long wavelength light therapy might have a role in preserving or improving our photoreceptors... and therefore, our vision. And indeed, that's what the data show.

Work by neuroscientist Glen Jeffery (mentioned at the start of this chapter) investigated whether direct viewing of long wavelength light could improve people's vision. In one study, he and his team enrolled twenty-four people between the ages of twenty-eight and seventy-two to look at a 670nanometer beam (red light) from a flashlight for three minutes per day for two weeks. Participants under the age of forty showed no change in vision. But those around forty years and Older showed a greater ability to discern colors at the end of the experiment. In a second study, the investigators used the same intervention but at a less intense level, meaning the light was not as close to the participants' eyes (though it was Still 670 nanometers, or red light) and found that to be suficient to boost vision for one week, again among those about forty years Old and over.9 The improvement in vision in these studies reached 20 percent among some participants, which is a significant and compelling result.

The researchers also observed a reduction in drusen, which are fat deposits that accumulate in the eyes as we age98 and that are connected

to age-related macular degeneration. Importantly, the second study by Jeffery and colleagues found that the timing of the long wavelength light therapy mattered. The benefit to vision occurred only if the red light was seen in the morning, likely because mitochondrial function changes across the twenty-four-hour cycle, with a particular sensitivity to light then. It is worth noting that the range of benefit was wide; some individuals experienced a greater boost than others. The underlying reasons of who benefits and who doesn't are not yet clear.

Studies ongoing in Glen Jeffery's lab are now focused on the use of long wavelength light therapy for advanced macular degeneration, retinitis pigmentosa, and glaucoma. To be clear: The most common causes of vision loss—cataracts, glaucoma, macular degeneration, and diabetes99 are unlikely to be caused by lack of long wavelength light exposure, although that possibility hasn't been ruled out either. All these forms of vision loss are typically associated with aging, and some forms have a Clear genetic contribution as well. Everyone should get an annual eye exam and get their eye pressures checked for glaucoma once a year. Do not rely on this protocol as the only way to treat or prevent these conditions.

That said, the ability for long wavelength light to slow, halt, or reverse vision loss is gaining more and more support. Along those lines, you might be wondering how exactly to adopt this protocol. Turns out it's very easy: Expose your face to long wavelength light from a concentrated red light source (not laser!) for five to ten minutes once or twice per week, and do it within three hours of waking. You don't need to put the light source terribly close to your eyes either; it's best you do it at a distance that is comfortable and doesn't force you to close your eyes. However, in this and other long wavelength light protocols, you are welcome to close your eyes.

It won't hinder the effect unless you wear dark occluding glasses— the ones that make it look like you're going into a tanning bed (which is UV by the way; I do not recommend those for reasons that ought to be obvious by now). Long wavelength light easily passes through your eyelids to reach your photoreceptors. You can get enough retinal long wavelength light exposure as a consequence of whole body long wavelength light illumination too. There is no evidence that more frequent or longer exposures than five to ten minutes once or twice per week are needed for the visual system benefits, but also no evidence that more would be harmful. Remember, the effects were only observed in people forty and Older and only if it was done within one to three hours of waking.

SOURCES TO CONSIDER: You have a lot of options when it comes to long wavelength light sources for photobiomodulation—for this use, and others. I Will discuss these in order from the least to the most costly. Option 1: The sun. The sun is rich with long wavelength light and it's Some people like LWL face masks or helmets for skin or hair, respectively. A review of the research on a phototherapy mask, published in the Journal of Clinical and Aesthetic Dermatology in 2018, concluded that this light therapy device was effective for treating acne and psoriasis, healing wounds, and reducing actinic keratosis, rough and scaly skin that can result from years of sun exposure.

I This study was not funded by the device manufacturer, but it was authored by the head of a private dermatology research institute rather than an academic laboratory research group. That said, there is no reason to think the results with a mask would be any different from the results with a light panel, except that the light is hitting a smaller area of skin. I think the panels are the most versatile as you can expose whatever body areas you want but obviously cost compare and decide for yourself. After deciding what is affordable for you, the most important factor in selecting a LWL device is that it exclusively emits long wavelength light of about 600 nm or longer. (We don't see infrared light, which is Why these are often called red light devices.) Generally speaking, the broader the range of long wavelengths that a device emits, the better the results. In other words, the most effective devices emit red, nearIR, and IR (600—850+ nm) light.

The Jeffery lab studies used red light at wavelengths of about 600 to 670 nanometers to produce the visual system benefits, while light at wavelengths of 850 nanometers and above penetrated into and through the body—which again, is a good thing. As another rule, LWL devices that feel warm to the skin at the distance you use them are going to be more effective, because that encourages more blood flow and lymphatic flow, and in doing so helps improve the anti-inflammation effects of long wavelength light. Some LWL devices also include non-UV short wavelength light, which can be helpful for pain relief. Most LWL devices use LEDS, but a few use incandescent bulbs. Lasers, which deliver a focused ray of light, tend to be higher in intensity —and they are brighter.

I don't recommend these for home use. Low-level laser therapy (LLLT) is a medical intervention and anyone using it to address aging or skin issues that are often treated with topical creams or other prescription interventions, such as wounds, acne, or other chronic skin conditions, should consult with a physician to ensure this approach is safe for you. As mentioned above, I strive to take briefwalks outside throughout the day

just before or after meals, and pace outside when on work calls, if I can. Obviously I wear clothing during these times, but even with most skin covered, I'm sure to get some long wavelength light exposure, especially on Clear days. I also use a JOOVV LWL panel for ten to thirty minutes once a day, generally three to five days per week.

The JOOVV panels are medical grade and emit red and IR light; note they are a Huberman Lab podcast sponsor. I also really like the LWL incandescent bulbs from SaunaSpace as they generate heat, and emit 600 to 1,000 nanometer light. They make actual red light saunas too using those same bulbs; they look like a small canvas tent with a large-bulb long wavelength light—incandescent panel inside, and they are terrific because they too generate a lot of heat—which is what you want for a sauna session—and have all the features of long wavelength light photobiomodulation. I confess I am not a fan of most other long wavelength light, or red light, saunas that use LEDs because they generally don't get hot enough to trigger the benefits of dry or wet traditional saunasm Dry heat saunas can have great benefits for cardiovascular health, for growth hormone production, and for enhancing red blood cell production leading to endurance performance enhancement, but those effects are generally seen using temperatures ranging from roughly 180 to 210 degrees Fahrenheit. Most red light saunas just don't get hot enough to create those benefits.

Rather than View any one LWL device as "the best," consider what you are able to access and afford, and use that. And remember the best and least expensive long wavelength light emitting source: good 01' sunlight.